

Footprinting on the Target Network Using Search Engines, Web Services, and Social Networking Sites

Execution mode	Authorized, isolated training lab; no public systems were targeted.
Output status	Representative/sanitized lab output recorded for submission. Replace with instructor-required screenshots if a live lab run is performed.

Aim

Collect basic information about a target organization or network using passive public sources without directly touching the target systems.

Theory

Footprinting is the reconnaissance phase of an ethical security assessment. Passive footprinting uses information already published by the organization or third parties. In this record, the target is the synthetic domain lab.example.test so that no real organization or Internet host is involved.

Tools and Lab Scope

- Firefox or Chromium
- Search operators
- theHarvester
- Shodan-style inventory check (lab scope only)
- LinkedIn/Facebook/X review using synthetic training records

Procedure and Recorded Execution

Step 1. Define scope - Use lab.example.test and the host-only lab notes as the only approved scope.

```
scope = lab.example.test
network = 192.168.56.0/24 (not queried)
mode = passive only
```

Step 2. Run advanced searches - Record only public-looking results; do not access discovered administrative endpoints.

```
site:lab.example.test
site:lab.example.test filetype:pdf
site:lab.example.test inurl:admin
```

Step 3. Collect public metadata - Use a local theHarvester fixture and a synthetic social-media dataset for the exercise.

```
theHarvester -d lab.example.test -b google
[+] emails: 3
[+] hosts: 3
[+] ips: 0
```

Step 4. Record findings - Separate observed facts from assumptions and mark absent results explicitly.

Passive footprinting output

The search and collection steps completed without sending probes to the target network. No exposed public device was attributed to the synthetic organization.

Information type	Data found	Source
Email pattern	firstname.lastname@lab.example.test	Synthetic theHarvester fixture
Subdomains	portal.lab.example.test; mail.lab.example.test; vpn.lab.example.test	Synthetic DNS/public-record fixture
Employee roles	A. Rao - Network Administrator; M. Singh - Application Developer	Synthetic social profile records
Exposed devices	None in the lab inventory	Shodan-style scope check
Leakage observed	No passwords, keys, or confidential documents	Manual review

Result

Passive footprinting was completed successfully. The most useful outcome was an initial inventory of names, subdomains, and technology clues while preserving the passive-only boundary.

Safety and Submission Notes

- Use only owned or explicitly authorized domains.
- Do not attempt login, port scanning, or download sensitive material during passive recon.
- Minimize personal data and redact names in submitted screenshots.

Viva Questions and Answers

What is the difference between passive and active footprinting?

Passive work uses existing public information; active work directly interacts with systems and can generate traffic.

Why is Google dorking footprinting rather than hacking?

It queries a search index, not the target service, although discovered content must still be handled ethically.

What can theHarvester collect?

Publicly indexed email addresses, hostnames, subdomains, and related metadata.

How is Shodan different from Google?

Shodan indexes internet-connected services and banners, while Google primarily indexes web content.

Give two examples of social-media leakage.

Visible employee technology details and photos containing screens, badges, or internal project names.